DATA3 PROMPT DOC — loader, plots, contours, and the physics push

This is the working task list for the DATA3/NF270 diafiltration analysis in the `dynamic-diafiltration-pyomo` repo (branch `refactored_unified_codebase`). It is written to be read **one card at a time** — every task is self-contained.

How to read this file (30 seconds)

- **Every task is one card** with the same five slots: **Goal** (one line) → **Why** → **Do** (checklist) → **Done when** → **Files**.
- **Status chips:** ✅ = done · 🔶 = partly done · ⬜ = not started. The dashboard below tells you where everything stands at a glance.
- **Part A** (Tasks 1–11) = the original loader/plots/contours work. **Part B** (Tasks 12–23) = added 2026-07-22. Task 11+k = point k of the 2026-07-22 request, so the numbering maps 1:1 to that message.
- Line numbers drift. **Skim the referenced file and confirm names/lines before editing.**

Status dashboard

#	Task (short name)	Status	Output / where it lives
1	Config-driven DATA2 time correction	✅	`apply_campaign_time_correction` in the library; `Architecture.md` note
2	Legend-off-data + shared DATA2 style	🔶	applied in new figures; no single reusable `DATA2_STYLE` helper yet
3	Per-experiment prediction figures (8 plots)	🔶	5-panel combined version in library (`plot_meas_vs_pred_conductivity`); 8 separate files + ΔP plot pending
4	Per-sheet concentration-range table	⬜	(`concentration_range.png` exists as a one-off)
5	7-channel σ×B WSSE slice contours	✅	`_run_Bsigma_channels.py` → `bform_study/Bsigma_channels/`
6	Rejection R vs log Pe (σ-family)	⬜	superseded/extended by Task 14
7	Fix S0/S from recorded feed rates	⬜	—
8	Square slice contours: all B-forms × all sheets	🔶	`_run_Bsigma_coeffs.py` → `bform_study/Bsigma_coeffs/` + A4 PDFs (NaCl set + representatives done)

#	Task (short name)	Status	Output / where it lives
9	Profile-likelihood contours (rigorous)	🔶	`_run_profile_contours.py` machinery; full campaign + LR regions pending
10	Units audit + unit-change sensitivity	⬜	—
11	Selectable cond ↔ conc converter (3 backends)	✅	`concentration_to_conductivity` / `conductivity_to_concentration`, runfile branch `v`
12	Pull Alex's regression repo + notes PDF	🔶	imported → `data3_regression_july26_AD/`; the *reading/reconciling* is pending
13	Consolidate ALL plotting into the library	🔶	meas-vs-pred ported (the worked example); contour drivers et al. pending
14	B(c_int) vs log Pe rejection plots (σ-trajectories)	⬜	—
15	Derive B–c_int correlation per basis function	⬜	—
16	The "beyond-Pe" evidence plot (6 representative runs)	⬜	—
17	Derive the replacement dimensionless number	⬜	—
18	Analysis with the new number × B(c_int)	⬜	—
19	Js/Jw vs Pe and vs the new number (2×2)	⬜	—
20	Electromigration: hammer vs scalpel	⬜	—
21	Diffusivity-coefficient audit (cations/anions)	⬜	—
22	Integrate the BoE coupled-diffusivity equation	🔶	Box archive imported → `box_multicomponent_diafiltration/`; the port is pending
23	Sensitivity-analysis plots as library functions	⬜	(`sigma_sensitivity-*.png` exist as one-offs)

Suggested order for Part B (cheap → heavy): 1. **Task 12** (read Alex's notes — it reframes everything else), 2. **Tasks 20 + 21** (paper-and-pencil physics + a value audit; no big compute), 3. **Tasks 16 → 17 → 18 → 19** (the Pe story, in that order — each feeds the next), 4. **Tasks 14 + 15** (rejection plots + derivations), 5. **Tasks 13 + 23** (library consolidation + sensitivity functions), 6. then the heavy campaigns: Tasks 7, 4, 10, 8, 9.

Ground rules (never violate)

1. **Do NOT modify any MATLAB `.m` file.** (`load_data.m` even has unresolved merge-conflict markers — read for the convention only.)
2. **Do NOT change DATA1 or DATA2 behavior.** Everything new is opt-in / config-driven; existing campaigns stay byte-for-byte identical.
3. Match the **DATA2 plotting convention** (colors + markers) — never invent a palette.
4. **Verify, don't assume:** keep the DATA1/DATA2 pytest suite green and visually inspect every new/changed figure.
5. Do not edit `conductivity_paper.py` — wrap it from the library instead.
6. `box_multicomponent_diafiltration/` and `data3_regression_july26_AD/` are **read-only archives** — copy out what you need.

Key files & environments

- **Core library (put reusable code HERE):** `refactored_codes_v1/refactored_ucb_library.py` — loader `_load_legacy_data_stru_from_excel`, solver `solve_model(...)` (`sim_opt=True` = square forward solve, `False` = fit), time correction `apply_campaign_time_correction`, converters `concentration_to_conductivity` / `conductivity_to_concentration`, meas-vs-pred `plot_meas_vs_pred_conductivity` + `run_nf270_meas_vs_pred`.
- **Interactive driver:** `refactored_codes_v1/refactored_ucb_runfile.py` (branch letters; `v` = meas-vs-pred is the wiring pattern to copy).
- **Conductivity physics (do not edit):** `refactored_codes_v1/conductivity_paper.py` (`variant_shedlovsky`, `msa_transport`) + `CONDUCTIVITY_PAPER_EXPLAINER.md`.
- **Contour drivers (to be absorbed by Task 13):** `_run_Bsigma_channels.py`, `_run_Bsigma_coeffs.py`, `_run_Bsigma_fixedLp.py`, `_run_profile_contours.py`, `_run_all_pairs_5channel.py`, `_run_all_11_sheets_5channel.py`.
- **Figure output root:** `UnifiedFramework/DATA3/results/paper_artifacts/nf270/`.
- **Alex's regression snapshot:** `data3_regression_july26_AD/` (notes = `DATA3_regression_notes_AD_2026-07-07.pdf`; see its `PROVENANCE.md`).
- **Box archive:** `box_multicomponent_diafiltration/` (BoE derivation + MATLAB; see its `PROVENANCE.md`).
- **Envs:** `~/miniforge3/envs/my-idaes-env/bin/python` (pyomo + IPOPT — anything that solves) · `~/miniforge3/bin/python` (plot-only).
- **DATA2 time-correction reference (read-only):** `legacy/data1_matlab/functions/load_data.m` + DATA2 notebooks.

PART A — original tasks (1–11)

Task 1 — Config-driven DATA2 time correction ✅ DONE

Goal. One reusable, campaign-level routine for the DATA2 permeate time correction — no more hand-patching per run.

What was built. - `apply_campaign_time_correction(data_stru, convention=None, *, workflow_family="DATA3")` in the library (+ helper `_permeate_anchor_index`). - Conventions: `"vial_close"` (default; byte-identical to the old inline logic), `"tube_transit"` (retired V_tube model), `"none"`. - Config: global `CAMPAIGN_TIME_CORRECTION = {"DATA3": "vial_close"}` + per-run override `data_config['permeate_time_correction_convention']`. - The Excel loader calls it automatically and stamps `data_stru["permeate_time_correction"]` (method, anchor, streams). - DATA1/DATA2 (`.mat`) never reach it → untouched. `Architecture.md` documents the knob ("Permeate time-correction convention").

Verified by. DATA1 smoke + DATA2 paper-comparison pytest green; DATA3 sheets load with the correction on via config.

Task 2 — Legend never covers data + one shared DATA2 style 🔶 PARTLY DONE

Goal. A single reusable styling helper so every figure uses the DATA2 palette and the legend never occludes the trends.

Why. Legends were sitting on top of the curves; colors were re-derived ad hoc per script.

Done so far. Legend-off-data placement applied to the coefficient-contour and meas-vs-pred figures. The DATA2 palette is encoded inside `plot_meas_vs_pred_conductivity`: mass `r.` (meas) / `b-` (model); retentate `ms` (meas) / `g-` (model); permeate vial `cs` + `r^` (cV) / `r-` (cH); backends by line style (— Shedlovsky, -- MSA, ·· DATA2).

Do (remaining). - [ ] Extract ONE shared `DATA2_STYLE` dict (colors + markers per stream) into the library and route the existing plots through it. - [ ] A small `place_legend_clear(ax)` helper (outside axes or emptiest corner).

Done when. A representative mass + concentration figure regenerates with the shared dict; nothing overlaps the data; colors match DATA2 exactly.

Task 3 — Per-experiment prediction-figure set (8 separate plots) 🔶 PARTLY DONE

Goal. For each DATA3 sheet: **8 separate** measurement-vs-prediction time-series figures (retentate and permeate always split), DATA2-styled, on the DATA2-corrected time axis.

1. Mass per vial (keep as-is — per-vial reset)
2. ICP retentate concentration
3. ICP permeate concentration
4. Conductivity probe — retentate (raw trace)
5. Conductivity probe — permeate
6. Conductivity-derived concentration — retentate (via Task-11 converter, per point)
7. Conductivity-derived concentration — permeate
8. Applied pressure ΔP (control input)

Done so far. The library's `plot_meas_vs_pred_conductivity` + `run_nf270_meas_vs_pred` (runfile branch `v`) produce a combined **5-panel** version of plots 1/4/5/6/7 with all three converter backends overlaid, with the full DATA2 time convention (shared `t_delay` origin; model drawn only for collection vials `i ≥ n_v0−1` — the startup-holdup vial is *not* predicted; per-vial mass reset; permeate ICP anchored at vial close).

Do (remaining). - [ ] Emit the **8 individual figure files** per sheet (add the ICP-only panels and the ΔP plot), reusing the same library internals — do it inside the library, not a new script (Task 13 rule). - [ ] Keep the combined panel as the cheap overview.

Done when. All 8 files render for one representative sheet (e.g. a concentrating $CaCl_2$ run), legend-clear, DATA2-styled, holdup handled.

Task 4 — Per-sheet concentration-range table ⬜ NOT STARTED

Goal. One table per sheet + one campaign summary: **min–max [mM]** of diafiltrate `c_D`, feed/retentate `c_F`, interfacial `c_in`, permeate `c_H`.

Why. The B(c) story needs to know what concentration window each experiment actually explores.

Do. - [ ] `c_D`, `c_F`, `c_H` from the loaded data; `c_in` from the fitted forward sim (`solve_model(...) → sim[vial]["cIn"]`) — label measured vs model. - [ ] Emit per-sheet CSV **and** markdown, plus one combined campaign table (a row per sheet).

Done when. Every DATA3 sheet has a table; spot-check `c_in > c_F` while concentrating and `c_in < c_F` while diluting.

Task 5 — 7-channel σ×B WSSE slice contours ✅ DONE

Goal. Extend the σ×B log-WSSE contour (fixed L_p, square forward solve at every node — the *slice*, NOT profile likelihood) to all seven response channels, one contour each.

What was built. `_run_Bsigma_channels.py` (`exp <rid>` / `all` / `plot <rid>`): mass · ICP retentate · ICP permeate · conductivity-probe ret/perm (σ-space) · cond-derived concentration ret/perm (c-space). ΔP excluded (control input). Reuses the `Bsigma_fixedLp_nofloor` grid; channels go through the Task-11 converter. **Gotcha handled:** the permeate-ICP channel needs absolute measurement-error floors (`FLOOR_MM=0.5`, `FLOOR_US=50`) or the near-zero startup-vial ICP (~0.08 mM) dominates the WSSE.

Output. `bform_study/Bsigma_channels/<rid>/single/{contour7.png, contourdata-7ch.csv, nodes.jsonl, grid.json}` for the three representative sheets ($CaCl_2$ MC2.05.07.24, $LaCl_3$ MC2.05.21.24, NaCl MC2.05.07.24).

Task 6 — Rejection R vs log Pe with the σ-family (extended by Task 14)

Goal. Analysis-phase figure: Spiegler–Kedem rejection curves `R(Pe; σ) = σ(1 − e^−Pe) / (1 − σ·e^−Pe)` for a grid of σ (0.1…0.99) vs log Pe, with each experiment's operating point overlaid — read the best-supported σ off the figure.

Why. Visualizes *why* σ is weakly identified: at large operating Pe all σ-curves sit at their plateau `R → σ`, so tiny R noise moves σ a lot.

Do. See **Task 14**, which wraps this and the B(c_int)-vs-log-Pe variant into one deliverable (same machinery, one card).

Task 7 — Fix S0/S from recorded feed rates (DATA3 lag mode) NOT STARTED

Goal. Stop guessing the apparatus feed terms: set `S0` (g/s) and `S` (g/hr) from the **recorded per-run values**, as fixed Params — not seeded guesses, not a free Var.

Why. Today `S0` is seeded from an `M_O`-based guess and `S` is a free `Var` (~lines 1810, 1956–58; lag balance `ode_mF_rule` ~line 2029: `dmF/dt = −S0 − area·ρ·Jw` on active legs, `dmF/dt = S/3600` on the closed leg). That's one phantom degree of freedom — and unconstrained on runs with no closed leg. This is a correctness cleanup, **not** a σ-identifiability fix (σ stays flat for the high-Pe-plateau reason).

Do. - [ ] Ingest recorded rates into `data_config` under explicit keys (`feed_rate_S0_g_per_s`, `feed_rate_S_g_per_hr`). - [ ] In `model_construct_inter`, DATA3-lag path only (`workflow_family == "DATA3"`): make BOTH `m.S0` and `m.S` fixed Params from those values. DATA1/DATA2/Overflow branches byte-identical. - [ ] Fall back to current behavior when a recorded value is missing; stamp the source (recorded vs fallback) on `data_stru` metadata.

Done when. A DATA3 lag sheet fits with S0/S fixed (metadata shows source); free-parameter count drops by one; DATA1/DATA2 byte-identical; pytest green. Expect fitted L_p/B/σ ≈ unchanged (mass channel already pinned S).

Task 8 — Square slice contours: every B(c) form × every sheet 🔶 PARTLY DONE

Goal. For each DATA3 single-salt sheet × each B(c) form (constant / linear / quadratic / cubic / saturating-exp / donnan): WSSE contours for the Task-5 channels, every node a **square forward solve** (no per-node optimization).

The decision rule that keeps every node square. 1. Fit each form once (or read `result_<form>.json`) → θ*. 2. Pin across the grid: `Lp` (optimum), `S0`/`S` (recorded — Task 7), and every B-coefficient NOT on an axis, at θ*. 3. Axes = σ × one B-parameter → zero free parameters → forward-solve the DAE and *evaluate* WSSE. 4. Label it a **conditional slice through the optimum** — its minimum coincides with the fit; it is *not* a profile.

Done so far. `_run_Bsigma_coeffs.py`: per-coefficient (β_k × σ) slices for linear/quad/cubic/sat (B built on interfacial conc `cIn`, `B_form_rule3`), subprocess worker + watchdog + JSONL resume + a cheap B(cIn)-physicality pre-filter; $\beta_0 \times \beta_1$ correlation slice for NaCl; A4 multi-page PDFs (`<rid>_coeff_contours_A4.pdf`). Library toggle **`NF270_BFORM_SLICE_UNBOUNDED_B`** (default False = byte-identical) because the poly `sim_opt` `m.B` is a bounded Var that's infeasible off-optimum. **Honest findings:** slices are feasible only near each optimum (off-optimum B(c) leaves the physical region → infeasible holes, themselves informative); **cubic is essentially un-sliceable for NaCl** (fitted B(cIn) goes negative over the run) but fine for $CaCl_2$/$LaCl_3$.

Do (remaining). - [ ] Extend to **all** DATA3 single-salt sheets × **all six** forms × all 7 channels; save under `bform_study/` with a small index. - [ ] Optional robustness: re-slice at MLE ± SE of the pinned coefficients.

Done when. Every (sheet, form, channel) has a slice; each minimum sits at the fitted (σ, B); the flat-σ pattern is consistent across forms and salts.

Task 9 — Profile-likelihood contours (rigorous counterpart) 🔶 PARTLY DONE

Goal. Same scope as Task 8, but at each node **re-optimize the non-axis coefficients** (nuisances λ): `Φ_profile(ψ) = min_λ WSSE(ψ, λ | Lp, S0, S fixed)` — the surface that supports likelihood-ratio confidence regions `CR = {ψ : Φ_profile(ψ) − Φ_min ≤ χ² threshold}`.

Why. The Task-8 slice is conditional — NOT a valid confidence region. The profile is the honest identifiability statement. By-product: the profiled trajectories `λ*(ψ)` directly visualize parameter correlations (β_0–β_1 trade-off; whether β_1 tracks σ).

Done so far. `_run_profile_contours.py` has the `method="profile"` campaign machinery (warm-start continuation from converged neighbours, per-node IPOPT CPU cap, `skip_sim_init` / bounded CasADi init, watchdog kill-restart, JSONL resume); earlier per-form profile contours exist for representative sheets.

Do (remaining). - [ ] Full campaign: every sheet × form × channel, with LR confidence regions overlaid (+ optionally the λ*(ψ) trajectories). - [ ] Note differences vs the Task-8 slice. (Task 7 shrinks λ → faster nodes.)

Done when. Profile surfaces exist per (sheet, form, channel); minima coincide with fits; LR regions drawn; flat-σ confirmed **rigorously**.

Task 10 — Units audit + unit-change sensitivity test ⬜ NOT STARTED

Goal. Prove the pipeline is dimensionally consistent; document every unit; add a test that re-expresses inputs in different units and demands identical physics.

Part A — audit. - [ ] Units table for every quantity: mass g · `S0` g/s · `S` g/hr · `cF/cV/cD/cIn/cH` mM · `Jw` (LMH vs cm/s — state which where) · `Lp` $L \cdot m^{-2} \cdot h^{-1} \cdot bar^{-1}$ · ΔP bar · Δπ bar (van 't Hoff `n_i·R·T·Δc`) · `B` · σ (–) · `D` cm^2/s · `k` cm/s · Pe (–) · `Am`, ρ, time (min vs s) · conductivity mS/cm vs µS/cm, M vs mM. - [ ] Term-by-term checks at the risky interfaces: `Jw` as velocity in `exp(Jw/k)` vs mass rate in `Am·ρ·Jw` vs `Pe = Jw·L/D`; **B's two conventions** (`Js = B·Δc` for single vs `Js = Jw·B·Δc` for the B(c) forms — different units; the `×Jw·1e4` apparent-µm/s conversion); ΔP vs Δπ pressure units; the `S/3600`; `(TF−TI)/Tauf` dimensionless; Shedlovsky/MSA input/output units + EC25 compensation. - [ ] Fix genuine mismatches DATA3-guarded; never silently change DATA1/DATA2.

Part B — sensitivity test. - [ ] Re-run one fit with an input in different units + correct conversion (ΔP in Pa with Lp converted; conc in M; SI area). Fitted Lp/B/σ and WSSE **must be invariant** — any drift is a hidden hard-coded unit (a bug). - [ ] A dimensional-scaling check (multiply input by known factor → outputs scale exactly as predicted). Wire 1–2 as DATA3-guarded pytest invariants.

Done when. `UNITS.md` (or an `Architecture.md` section) with the table + report; invariance test passes on ≥1 sheet.

Task 11 — Selectable conductivity ↔ concentration converter ✅ DONE

Goal. One converter pair with a selectable backend so every conductivity-derived quantity can be computed under — and compared across — three models.

What was built. In the library (wrapping `conductivity_paper.py`, unedited): - `concentration_to_conductivity(c_mM, salt, T=25, *, method=...)` and `conductivity_to_concentration(cond_uS, salt, T=25, *, method=...)`, `method ∈ {"shedlovsky", "msa", "data2"}`. - `"shedlovsky"` = `variant_shedlovsky` (inverse via per-point bisection; NaN-safe where the map folds at high conc); `"msa"` = `msa_transport` (inverse NaN at very low conc — bisection limitation → permeate-side gaps); `"data2"` = the DATA2 linear calibration `conc[mM] = 0.008813·cond[µS/cm] − 0.6949` (`DATA2_CONDUCTIVITY_CALIBRATION`). - Round-trip exact for shedlovsky/data2; all conductivity channels (Tasks 3/5/8/9) route through these.

Key finding. The three backends **agree closely** on this campaign → the probe-vs-ICP concentration gap is calibration/model, *not* the conversion equation.

PART B — new tasks (12–23), added 2026-07-22

(Task 11+k = point k of the 2026-07-22 request.)

Task 12 — Pull in Alex's regression work + his notes PDF 🔶 IMPORT DONE, READING PENDING

Goal. Alex (Dowling, PI) built a regression analysis on a **different process model** — an extended **Nernst–Planck / Donnan-equilibrium** membrane description regressing the membrane **fixed charge χ** and a thermodynamic **partition factor δ*** — physics our DATA2-style Spiegler–Kedem (Lp, B, σ) workflow does not have. Bring it into this branch and reconcile it with ours.

Already done (2026-07-22). - Repo snapshot imported: `data3_regression_july26_AD/` (from https://github.com/dowlinglab/data3_regression, branch `main`, commit `5b8b1b7` of 2026-07-17, `.git` stripped; see `PROVENANCE.md` there). - His write-up imported and renamed: `data3_regression_july26_AD/DATA3_regression_notes_AD_2026-07-07.pdf` ("Towards Multicomponent Diafiltration: Mathematical Modeling and Regression Notes", 62 pp). ⚠ The repo HEAD is 10 days **newer** than the PDF ("Added feedback from meeting with collaborators") — trust the repo where they disagree.

Do (the real work). - [ ] Read the notes; map his model onto ours: χ, δ* (Donnan) ↔ our B(c_int), σ; his CP-corrected `c_int` ↔ our film-model `cIn`. - [ ] His statistics are directly reusable on OUR fits: autocorrelation-corrected confidence regions (i.i.d. CIs are ~2.5–2.7× too narrow on dense time series — this hits our WSSE contours too), effective sample size n_eff, profile likelihood, bootstrap, raw-measurement regression (§2.3–2.4 of the notes). - [ ] Walk his §1.2 "open decisions" list; flag which ones touch our workflow. - [ ] Short reconciliation note (md) in `data3_regression_july26_AD/` or `Architecture.md`: what we adopt, what stays separate.

Done when. The reconciliation note exists and at least the autocorrelation/n_eff correction has a concrete plan (or implementation) for our contour/CI story.

Task 13 — Consolidate ALL plotting into `refactored_ucb_library.py` 🔶 PATTERN ESTABLISHED

Goal. Every plotting capability developed in standalone `refactored_codes_v1/_*.py` scripts becomes a **library function** (+ a runfile branch where it's a per-sheet campaign). **Especially the 5-panel contour plots that are our norm for DATA3 single-salt sheets.** No orphan scripts. *(This absorbs the scattered "new module or library helper" phrasing that was in Tasks 2/3 — this card is now the single consolidation task.)*

The worked example to copy (already done). meas-vs-pred: `plot_meas_vs_pred_conductivity` (figure fn) + `run_nf270_meas_vs_pred` (per-sheet batch: load → fit → forward-sim → plot) + runfile branch letter `v` + the standalone `_make_meas_pred_nacl.py` **deleted** after the port.

Do — port these, one at a time, same pattern (figure fn + `run_nf270_*` dispatcher + runfile letter + delete/thin the script): - [ ] The **5-channel σ×B contour norm** (`_run_all_pairs_5channel.py`, `_run_all_11_sheets_5channel.py`, `_run_5channel_demo.py`, `_contour_batch_sigma_B.py`). - [ ] The **7-channel** slice contours (`_run_Bsigma_channels.py`, Task 5). - [ ] The **coefficient contours + A4 renderer** (`_run_Bsigma_coeffs.py`, Task 8) — keep the watchdog/resume machinery; it can live in the library too. - [ ] Fixed-Lp contours (`_run_Bsigma_fixedLp.py`) and profile contours (`_run_profile_contours.py`, Task 9). - [ ] Prediction generator (`_make_predictions.py`) → becomes the Task-3 8-figure set. - [ ] The stirred-cell mass-balance / interfacial-conc time plot (outputs in `bform_study/retentate_massbalance/`). - [ ] Anything else `_make_*.py` / `_run_*.py` that renders a figure the campaign should be able to regenerate.

Rules. Solver-touching code guarded `workflow_family="DATA3"`; heavy campaigns get a `run_nf270_*` batch entry point; scripts that remain (if any) are thin CLI shims that import the library. `Architecture.md` gets one line per new branch letter.

Done when. The runfile can regenerate every figure class end-to-end with no standalone script logic; pytest green; deleted scripts leave no importers.

Task 14 — Rejection plots: B(c_int) vs log Pe with σ-trajectories ⬜ NOT STARTED

(extends Task 6; do both variants with the same machinery)

Goal. Per experiment: plot **B(c_interfacial) on Y vs log Pe on X**, with **one trajectory per σ value**, so the σ-consistency of each run is visible.

Careful — there are TWO Peclet numbers in play; label which one: - film/CP Peclet `Pe_film = Jw/k` (drives `c_in`), and - membrane Peclet `Pe_m = Jw(1−σ)/B` (drives rejection). For the σ-trajectories on a B-vs-Pe_m plot, note `B = Jw(1−σ)/Pe_m` — at a given instantaneous `Jw`, each σ is a straight line of slope −1 in log B–log Pe space; the run's per-vial (Pe, B(c_int)) points trace a trajectory across them as `c_int` (and `Jw`) evolve.

Do. - [ ] Per sheet: compute per-vial `Jw`, `c_int` (from the fitted forward sim), `B(c_int)` for the fitted form, both Peclets; plot B vs log Pe with the σ-family overlaid (σ grid 0.1…0.99); mark time/vial order (arrow or colorbar) so the trajectory reads as a path. - [ ] Also render the **Task-6 classic**: `R(Pe; σ) = σ(1−e^−Pe)/(1−σe^−Pe)` σ-family vs log Pe with the observed per-vial rejection `R_obs = 1 − c_p/c_F` overlaid — best-supported σ read off the figure. - [ ] **Suggested extra (pick up if useful):** the most physically direct variant is **B(c_int) vs c_int** (it IS the correlation we fit) with vial order marked, and `R_obs` vs `Jw` as the operating-curve companion. Offer all; recommend the (a) R-vs-logPe + (b) B-vs-logPe pair as the deliverable.

Done when. Both figures render per single-salt sheet (library functions — Task 13 pattern), σ-family legend clear of data, Peclet definition stated on the axis label.

Task 15 — Derive the B–c_int correlation for each basis function NOT STARTED

Goal. A short derivation (as in the DATA2 paper, `UnifiedFramework/DATA3/published_works/DATA2_main.pdf`) of what each basis we fit — constant, linear, quadratic, cubic, saturating-exponential, Donnan — **implies physically** for B(c_interfacial), and what each coefficient means.

Why. Right now the forms are curve-fitting choices. The DATA2 paper's route (solution-diffusion permeability + concentration-dependent partitioning, Taylor expansion about the operating window) turns each β_k into a statement about partition-coefficient derivatives — that's what makes the AIC/order-selection result mean something.

Do. - [ ] Follow the DATA2 derivation: start from `Js = B(c)·(c_in − c_H)` with `B = D_m·K(c)/L` (solution–diffusion), expand `K(c)` (Taylor for the polynomial family; Langmuir-type saturation for sat-exp; Donnan exclusion `K ∝ (c/χ)^{|z_co|/…}`-style for the charged form). - [ ] Map: β_0 = dilute-limit permeability; $\beta_1 \propto \partial K/\partial c$; $\beta_2 \propto \partial^2 K/\partial c^2$; sat-exp amplitude/scale = site saturation; Donnan exponent = valence/charge screening. - [ ] Write it as a 2–3 page note (md or LaTeX; native equations), with the DATA2 equations cited by number; cross-link to the AIC order-selection result and to Alex's Donnan mechanistic model (Task 12) — his χ, δ* should appear as the parameters of the Donnan-form K(c).

Done when. The note exists, each fitted coefficient has a physical reading, and the Donnan-form derivation connects our B(c) to Alex's (χ, δ*).

Task 16 — The "beyond-Pe" evidence plot NOT STARTED

Goal. Pe-based (diffusion + convection only) reasoning assumes those are the only dominant effects. Our campaign suggests otherwise. Produce **the one plot that shows this with no room for doubt**, using one representative experiment per salt (NaCl, $CaCl_2$, $LaCl_3$) × per regime (diluting, concentrating) — 6 runs.

Candidate figures (build (a); (b) is the kill shot; (c) optional). - [ ] **(a) Constant-coefficient SK failure, per run:** fit each run with constant (Lp, B, σ); plot observed per-vial rejection vs log Pe on top of the fitted `R(Pe; σ)` curve, and (inset or second row) the residuals vs `c_int`. If diffusion+convection sufficed, residuals are noise; instead they trend systematically with `c_int` — same membrane, same salt, coefficient must drift with concentration. - [ ] **(b) Cross-salt collapse failure:** the 6 runs' `R_obs` vs log Pe on ONE axis. Pure diffusion–convection predicts each salt's points collapse onto a single σ-plateau curve (differences only via D in Pe). Show the valence ordering (NaCl > $CaCl_2$ > $LaCl_3$ rejection at comparable Pe) is far larger than the D-spread can produce — quantify: recompute Pe with each salt's D (Task 21 values) and show the curves STILL don't collapse. - [ ] (c) `B_apparent` per vial (from `Js/(c_in − c_H)`) vs `c_int` across the 6 runs — orders-of-magnitude drift at similar Pe = a "constant" that isn't.

Done when. The chosen figure(s) exist for the 6 representative runs with a caption that states the argument in two sentences; the D-spread quantification is in the caption or an SI table.

Task 17 — Derive the replacement dimensionless number NOT STARTED

Goal. If Pe (convection/diffusion) is not the whole story, derive — don't guess — the dimensionless group(s) that should join or replace it for our charged-membrane system.

Route. Nondimensionalize the **extended Nernst–Planck + Donnan** problem (the model Alex uses, Task 12; also `box_multicomponent_diafiltration/Manuscript/JAO/Transport, MembranePhase - Diffusion, Convection, Electromagration.docx`). The natural groups that fall out: - `Pe = Jw·δ/D` — convection vs diffusion (what we already use), - **`ξ = χ / (z·c_int)`** — membrane fixed charge vs external ionic strength: the Donnan/Teorell–Meyer–Sievers exclusion parameter. Controls co-ion exclusion, hence rejection, hence the *apparent* B(c). Large ξ = strong exclusion; ξ → 0 = neutral membrane, Pe story recovered. - `φ_D = z·F·Δψ_Donnan/(R·T)` — Donnan potential vs thermal voltage (equivalent information to ξ through the Donnan isotherm).

Recommendation to verify in the derivation: ξ is the right second axis — it is built from Alex's regressed χ and our model's `c_int`, it is per-vial-computable, and its c_int-dependence is exactly why B looks concentration-dependent.

Do. - [ ] 1–2 page derivation note: nondimensionalize, exhibit Pe and ξ (and φ_D), state the limits (ξ→0 neutral; Pe→∞ convective). - [ ] Library helper computing per-vial ξ (χ from Alex's per-salt fits; `c_int` from our forward sim), DATA3-guarded.

Done when. The note + helper exist; ξ values per vial are available for Tasks 18–19.

Task 18 — What does (new number) × B(c_int) tell us? ⬜ NOT STARTED

Goal. Re-do the Task-14 analysis with ξ in place of (or alongside) Pe and answer: **is it useful?**

The test is collapse. Plot `B(c_int)` (and `R_obs`) vs ξ for the 6 representative runs (and then all single-salt sheets): - If the salt-to-salt and regime-to-regime spread that Pe could **not** explain collapses onto one master curve in ξ — the charge physics is the missing driver, and ξ is the right correlating variable (answer: useful). - If not, quantify what remains (partitioning non-ideality δ^*, dielectric exclusion) and say so honestly.

Do. - [ ] B(c_int) vs ξ and R_obs vs ξ, per salt and pooled; same 6 representative runs as Task 16, then the full campaign. - [ ] Quantify collapse: compare a pooled fit's WSSE/R^2 in ξ vs in Pe vs in c_int; report the ranking.

Done when. The collapse comparison (ξ vs Pe vs c_int) is plotted + quantified, with a one-paragraph verdict on usefulness.

Task 19 — Js and Jw vs Pe and vs the new number (2×2) ⬜ NOT STARTED

Goal. Four panels per (representative) experiment set: `Js` vs Pe, `Jw` vs Pe, `Js` vs ξ, `Jw` vs ξ. State up front what each can and cannot show.

Expected information content (write this into the caption). - `Js` vs Pe — the diffusive→convective transition; curvature/slope reads the effective B; the most diagnostic of the four. - `Jw` vs Pe — **near-tautological** (Pe ∝ Jw when δ/D is ~constant): expect a straight line; only *deviations* from linearity are informative (they mean k or D changed — CP regime shift). Plot it as the control, say so. - `Js` vs ξ — charge regulation of solute flux: does Js drop as exclusion strengthens (ξ ↑ as c_int ↓)? This is the Donnan signature. - `Jw` vs ξ — should be ~flat unless osmotic/charge effects feed back on water flux ($\sigma \cdot \Delta\pi$ coupling); a trend here implicates the reflection term.

Do. - [ ] Per-vial Js (= d(m_V·c_V)/dt / area, or from the model), Jw, Pe, ξ; 2×2 figure per run + one pooled version colored by salt; library function (Task 13 pattern).

Done when. The 2×2 renders for the 6 representative runs + pooled; captions state the expected/observed information content; a short verdict on which panels carry signal.

Task 20 — Electromigration: hammer or scalpel? ⬜ NOT STARTED

Goal. You've been told explicit electromigration modeling is "a hammer for a job that needs a scalpel." **Is that true?** Give the simple physics-based answer with equations.

The answer to write up (verify each step). It is TRUE for single-salt fitting, FALSE-but-tractable for multicomponent: 1. Extended Nernst–Planck flux: `J_i = −D_i∇c_i − z_i c_i (D_i F/RT) ∇φ + c_i v`. 2. Zero net current: `Σ z_i J_i = 0` → solve for ∇φ and substitute back. 3. **Single binary electrolyte:** the migration term is eliminated *exactly*, leaving Fickian diffusion with the ambipolar (Nernst–Hartley) coefficient `D_± = (z_+ − z_−) D_+ D_− / (z_+ D_+ − z_− D_−)` — i.e. one scalar D per salt. Simulating the electric field explicitly adds **zero** physics for a single salt (the hammer); the scalpel is (a) using the correct ambipolar `D_±` (Task 21) and (b) Donnan partitioning at the interfaces (Alex's χ, δ*). 4. **Multicomponent (2 cations + common anion):** migration coupling does NOT vanish — but it reduces to a concentration-dependent **coupled diffusion matrix** $D_{ij}(c_1,c_2)$ (off-diagonals = one ion dragging another via the shared field). That is *exactly* what the BoE MATLAB does (`calculate_D` in `box_multicomponent_diafiltration/Analyses/MATLAB codes/BackoftheEnvelope_Calculations.m`, lines 98–122) — still no explicit field solve (still a scalpel, Task 22).

Do. - [ ] Write the 1-page note with the three equations above + the D_± values for $NaCl/CaCl_2/LaCl_3$ (from Task 21); cross-ref the JAO transport docx variants (diffusion / +convection / +electromigration) in the Box archive and Alex's notes §1.5/§1.8.

Done when. The note exists and gives the advisor-ready two-sentence answer: "for single salts, zero-current NP reduces exactly to Fick with D_± — explicit migration adds nothing; for mixtures it becomes a D_ij matrix, which is the BoE route, still no field solve."

Task 21 — Are our diffusivity coefficients right? ⬜ NOT STARTED

Goal. Audit every diffusion coefficient in our pipeline (film model `k`, Pe, any D in the loader/analysis): which value, which convention (single-ion vs salt/ambipolar), which source — and are they right for each cation/anion pair?

Reference values to audit against. - Per-ion (BoE, `BackoftheEnvelope_Calculations.m` lines 21–24, m²/s): `D_Na = 1.33e−9`, `D_Ca = 0.79e−9`, `D_La = 6.26e−10`, `D_Cl = 2.03e−9`. - Ambipolar salt values (Nernst–Hartley from those ions, 25 °C, infinite dilution): NaCl ≈ 1.61e−9, $CaCl_2$ ≈ 1.33e−9, $LaCl_3$ ≈ 1.29e−9 m²/s. **Watch for the classic error:** using the *cation* D for the *salt* (underestimates NaCl by ~20%, $LaCl_3$ by ~2×). - Concentration dependence: Robinson & Stokes chapters are in the Box archive (`Conductivity Analyses/`) if we need D(c) beyond infinite dilution.

Do. - [ ] Find every place a D (or k built from D) enters the library/loader; table of used-vs-literature values per salt, with convention labeled. - [ ] Check the mass-transfer correlation too — the BoE stirred-cell form is $k = 0.23 \cdot v_0^{0.57} \cdot D^{0.67} / (\nu^{0.24} \cdot b^{0.43})$ (`.m` lines 124–130); compare against what our film model uses. - [ ] Re-fit one sheet per salt with corrected D: report the effect on (Lp, B, σ) and on Pe (this feeds Tasks 14/16).

Done when. The audit table exists; any wrong value is fixed (DATA3-guarded) or justified; the sensitivity of the fits to the D choice is quantified.

Task 22 — Integrate the BoE coupled-diffusivity equation 🔶 ARCHIVE IMPORTED, PORT PENDING

Goal. Port the diffusivity-coefficient machinery derived in the Box “Back of the Envelope” documents into `refactored_ucb_library.py` as opt-in functions, so the multicomponent film model can use it.

Already done (2026-07-22). The Box folder is in the branch: `box_multicomponent_diafiltration/` (see its `PROVENANCE.md`). The key sources: - `Manuscript/Back of the envelope calculations for SI.docx` — the derivation (the boxed diffusivity-coefficient equation lives here). - `Analyses/MATLAB codes/BackoftheEnvelope_Calculations.m` — reference implementation: `calculate_D(c1,c2)` builds the 2×2 coupled NP diffusion matrix `D_ij` (two cations + common anion, zero current), then the film problem is solved by eigen-decomposition (`Δc_a = T·diag(exp(Jw·δ/σ_eig))·T⁻¹·Δc_0`) → interfacial concentrations. - `Analyses/Back of the Envelope Calculations_Coefficients Calculations.pptx` + `BoE Analysis.xlsx` — the worked numbers.

Port carefully — known quirks in the `.m` (do NOT copy blindly, and do NOT edit the `.m` itself): - Line 48: `Delta_La = params.DCa_s/k(params.DLa_s)` — numerator uses **DCa** (looks like a typo for DLa). - `Delta_avg` averages Na/La/Cl only (sheet-specific: NaCl+$LaCl_3$ mixture); `calculate_D` hard-codes D1=Na, D2=La and `chi = 0` (placeholder).

Do. - [ ] Library functions (DATA3-guarded, opt-in): `boe_coupled_D_matrix(c1, c2, ions=...)` (general z_i, D_i — not hard-coded), `boe_stirred_cell_k(D)` (the $0.23 \cdot v_0$^0.57 correlation with rig constants b=0.0254 m, ν=1.003e−6 m²/s, 350 RPM as defaults), and the eigen-decomposed film solve → `c_int` per ion. - [ ] **Single-salt limit test:** with one salt, `D_ij` must collapse to the ambipolar D_± and the film solve to our current `exp(Jw/k)` CP model — pytest that. - [ ] Cross-validate against `BoE Analysis.xlsx` numbers for one mixture vial.

Done when. The functions exist + tests pass (single-salt reduction, xlsx cross-check); a mixture c_int can be computed end-to-end from the library.

Task 23 — Sensitivity-analysis plots as library functions ⬜ NOT STARTED

Goal. Standard sensitivity figures answering “are model + data sensitive to our parameters?” — as **functions in `refactored_ucb_library.py`** (Task 13 pattern), not scripts.

Do. - [ ] **Local, forward:** perturb each parameter (Lp, σ, each βₖ, S0/S, D, k) ±X% about the fit; forward-solve (square — no optimization); plot (a) trajectory ribbons (mass, cF, cV bands under the perturbation) and (b) a tornado chart of ΔWSSE per parameter (total + per response channel). - [ ] **Local, derivative-based:** reuse `calc_FIM` — report per-parameter sensitivities `∂y/∂θ` scaled, eigen-spectrum of the FIM (ties to the contour/identifiability story; cite the flat-σ finding). - [ ] Optional if cheap: a coarse Morris screening across all sheets. - [ ] Entry points: `plot_parameter_sensitivity(...)` + `run_nf270_sensitivity(...)` + a runfile branch letter; precedent one-offs (`sigma_sensitivity-*.png`) get regenerated through the new path.

Done when. One command produces the ribbon + tornado + FIM-spectrum set for a representative sheet (and batch for all sheets); figures land under `paper_artifacts/nf270/sensitivity/`; pytest green.

Verification checklist (run at the end of any working session)

- ☐ `pytest` DATA1 + DATA2 smoke/paper-comparison green (they stay untouched).
- ☐ Every new/changed figure rendered to PNG and **looked at** (legend clear, DATA2 colors, time correction visible in the mass panel).
- ☐ New library entry points reachable from the runfile; `Architecture.md` updated (branch letters, config knobs).
- ☐ Anything ported out of a standalone script: script deleted or thinned to a shim; no orphan importers.
- ☐ Heavy campaign outputs land under `UnifiedFramework/DATA3/results/paper_artifacts/nf270/` with a small index (JSON/md) per campaign.
- ☐ Derivation notes (Tasks 15/17/20) committed alongside the figures they explain.

Assumptions baked in (change if wrong)

- Task 3 = **8 separate figure files** per experiment (+ the combined panel as an overview).
- Single-salt conversions via `variant_shedlovsky` inversion remain the default; MSA is the multi-salt hook.
- ξ (dimensionless fixed charge, Task 17) is the working candidate for “the new dimensionless number” until the derivation says otherwise.
- The BoE port (Task 22) generalizes the hard-coded Na/La pair to arbitrary (z_i, D_i) — the `.m` stays untouched.